

Balancing Minimum Spanning Trees and Multiple-Source Minimum Routing Cost Spanning Trees on Metric Graphs

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Abstract

The building cost of a spanning tree is the sum of weights of the edges used to construct the spanning tree. The routing cost of a source vertex s on a spanning tree T is the total summation of distances between the source vertex s and all the vertices d in T . Given a source vertices set S , the multiple-source routing cost of a spanning tree T is the summation of the routing costs for source vertices in S . Both the building cost and the multiple-source routing cost are important considerations in construction of a network system. A spanning tree with minimum building cost among all spanning trees is called a minimum spanning tree (MST), and a spanning tree with minimum k -source routing cost among all spanning trees is called a k -source minimum routing cost spanning tree (k -MRCT). Usually a k -MRCT of a graph G with respect to k sources is not a MST of G , and vice versa. This paper proposes an algorithm to construct a spanning tree T for a metric graph G with a source vertex set S such that the building cost of T is less than $1 + 2/(\alpha - 1)$ times of that of a MST of G , and the k -source routing cost of T is less than $\alpha(1 + \frac{2(k-1)(n-2)}{k(n+k-2)})$ times of that of a k -MRCT of G with respect to S , where $\alpha > 1$, $k = |S|$ and n is the number of vertices of G .

1 Introduction

Two important problems in network design and many other fields are the minimum spanning tree (MST) problem and the k -source minimum routing cost spanning tree (k -MRCT) problem. Let $G = (V, E, w)$ be a connected and undirected

graph, where V , E and w denote the vertex set, the edge set and the nonnegative weight function on E , respectively. The MST problem is to find a spanning tree T of G such that the total length of the edges of T is a minimum one among all spanning trees of G . Given k vertices of V as sources and all vertices of V as destinations, the k -MRCT problem is to find a spanning tree T of G such that the total distance from sources to destinations on the tree T is minimized among all spanning trees of G .

Both the MST problem and the k -MRCT problem are widely studied problems. For the first problem, Kruskal's algorithm [5] runs in $O(|E| \log |V|)$ time and Prim's algorithm [6] runs in $O(|E| + |V| \log |V|)$ time. (See the article by Graham and Hell [2] for a history of MST problem.) Note that 1-MRCT problem is exactly the shortest path tree (SPT) problem and can be solved by Dijkstra's algorithm [1] in $O(|V|^2)$ time. Wu *et al.* [7] have shown that 2-MRCT problem is NP-hard, and proposed a polynomial time approximation scheme (PTAS) algorithm for it. The $|V|$ -MRCT problem has been proven by Johnson *et al.* [3] to be NP-hard, and Wu *et al.* [10] have proposed a PTAS algorithm for it.

In fact, the k -MRCT problem is a special case of the sum-requirement optimal communication spanning tree (SROCT) problem. Given a non-negative weight for each vertex of V , the weighted routing cost of a vertex is the product of the weight of the vertex multiplied by the total distance from the vertex to all the other vertices. The SROCT problem is to find a spanning tree T of G such that the total weighted routing cost of V on the tree T is a minimum one among all spanning trees of G . The k -MRCT problem is a special case of the SROCT problem in which the weight of each source vertex is one and the weights of the other vertices are zeros. Since some special cases of the SROCT problem are NP-hard, the SROCT problem is NP-hard. Wu *et al.* [8] have proposed a 2-approximation algorithm for it.

The balancing MST and k -MRCT problem,

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$\text{Balance}(x, y, k)$, is to find a spanning tree T of G such that (1) the total tree length of T is less than x times to the length of a minimum spanning tree and (2) the total k -source routing cost on T is less than y times to the optimal k -source routing cost among all possible spanning trees of G . In 1995, Khuller *et al.* [4] proposed algorithm Find-Last to solve the $\text{Balance}(1 + \frac{2}{\alpha-1}, \alpha, 1)$ problem in $O(n)$ time when a MST and a SPT of G are given, where $\alpha > 1$ and $n = |V|$. In 2002, Wu *et al.* [9] solved the $\text{Balance}(f(i) + \frac{2}{\alpha-1}, \frac{i+3}{i+1}\alpha, n)$ problem in $O(n^{i+1})$ time, where $\alpha > 1$ and $f(1) = 1$, $f(2) = 2$ and $f(i) = 3$ when $i > 3$. In this paper, we give an algorithm to solve the $\text{Balance}(1 + \frac{2}{\alpha-1}, \alpha(1 + \frac{2(k-1)(n-2)}{k(n+k-2)}), k)$ problem for a metric graph G in $O(kn)$ time when a MST of G and SPT's of G rooted at the k source vertices are given. These results are summarized in Table 1.

The remaining sections are organized as follows. Some basic definitions and notations are given in Section 2. The algorithm for the balancing MST and k -MRCT problem is presented in Section 3. Finally, we give the conclusions in Section.

2 Basic Definitions and Notations

In this section, we give some basic definitions and notations that will be used in the discussion of the following sections. Let $G = (V, E, w)$ denote a connected graph with nonnegative edge weights, where V , E and w denote the vertex set, the edge set and the nonnegative weight function on edges, respectively. First of all, we give the following definitions:

1. A spanning tree $T = (V_T, E_T, w_T)$ of $G = (V, E, w)$ is a tree that connect all vertices of G ; i.e. $V_T = V$, $E_T \in E$, and $w_T(e) = w(e)$ for all $e \in E_T$.
2. The building cost of a graph $G = (V, E, w)$, denoted by C_G , is the summation of the weights of the edges of G ; i.e. $C_G = \sum_{e \in E} w(e)$.
3. The distance for a pair of vertices $s, d \in V$ on a graph G , denoted by $D_G(s, d)$, is the summation of the weights of the edges on the shortest path from s to d on G .
4. A minimum spanning tree of G , denoted by $\text{MST}(G)$, is a spanning tree with minimum building cost among all spanning trees of G . A shortest path tree of G rooted at r , denoted

by $\text{SPT}(G, r)$, is a spanning tree of G such that $D_G(r, v) = D_{\text{SPT}(G, r)}(r, v)$ for any vertex v in G .

5. The routing cost of a source vertex $s \in V$ on $G = (V, E, w)$ is the summation of the distances from s to all the vertices $d \in V$. The routing cost of a set S of source vertices on G , denoted by $R_G(S)$, is the summation of the routing costs for all the source vertices in S ; i.e. $R_G(S) = \sum_{s \in S} \sum_{d \in V} D_G(s, d)$.
6. A minimum routing cost spanning tree of G for a set S of source vertices, denoted by $\mathcal{T}_G(S)$, is a spanning tree of G with minimum routing cost among all spanning trees of G .
7. Graph $G = (V, E, w)$ is a *metric graph* if and only if G is a complete graph in which $D_G(p, q) \geq 0$ and $D_G(p, q) + D_G(q, r) \geq D_G(p, r)$ for all $p \neq q \neq r \in V$.

3 Balancing MSTs and k -MRCTs on Metric Graphs

In this section, we present a method to construct a spanning tree that balances the building cost and the k -source routing cost for a given metric graph with k source vertices. Recall that algorithm Find-Last in [4] can find out a spanning tree that balances the building cost and the single-source-all-destination shortest path for a given graph with a source vertex. We use Algorithm Find-Last(G, s, α, T_M, T_S) to denote the algorithm proposed in [4] by Khuller, and the input and output of this algorithm are described as follows:

Algorithm Find-Last(G, s, α, T_M, T_S)
Input: metric graph $G = (V, E, w)$ with a source vertex s of V , $\alpha > 1$, $\text{MST}(G)$ T_M , $\text{SPT}(G, s)$ T_S
Output: a spanning tree T of G

The following lemma states the performance about a tree produced by Find-Last.

Lemma 1 [4] Algorithm Find-Last(G, s, α, T_M, T_S) finds out a spanning tree T of G in $O(n)$ time, where $n = |V|$, such that
(1) $C_T \leq (1 + \frac{2}{\alpha-1}) C_{\text{MST}(G)}$, and
(2) $D_T(s, d) \leq \alpha D_G(s, d)$ for all $d \in V$.

Algorithm 1 described in Figure 1 is to find out a spanning tree for balancing MST and k -MRCT.

Problem	Time	Source	Remark
Balance($1 + \frac{2}{\alpha-1}, \alpha, 1$)	$O(n)$	[4]	metric graphs
Balance($f(i) + \frac{2}{\alpha-1}, \frac{i+3}{i+1}\alpha, n$)	$O(n^{i+1})$	[9]	metric graphs
Balance($1 + \frac{2}{\alpha-1}, \alpha(1 + \frac{2(k-1)(n-2)}{k(n+k-2)}), k$)	$O(kn)$	This paper	metric graphs

Table 1: The performance ratios and running times of related results for the balancing MST and k -MRC_T problem, where $\alpha > 1$, $n = |V|$, $f(1) = 1$, $f(2) = 2$ and $f(i) = 3$ for $i > 3$.

First of all, the algorithm finds a spanning tree T_i for each source vertex s_i by Find-Last, and then output the tree with minimum routing cost. Since algorithm Find-Last needs $O(n)$ time, the total time of Algorithm 1 is $O(kn)$ time, where $n = |V|$ and $k = |S|$. The following lemma comes from Lemma 1 and Algorithm 1 immediately:

Lemma 2 If T is a tree found by Algorithm 1, then we have $C_T \leq (1 + \frac{2}{\alpha-1})C_{\text{MST}(G)}$.

Lemma 3 If T is a tree found by Algorithm 1, then we have

$$R_T(S) \leq \alpha(1 + \frac{2(k-1)(n-2)}{k(n+k-2)})R_{T_G(S)}(S)$$

where $n = |V|$ and $k = |S|$.

Proof:

According to Algorithm 1, T is one of T_i with the minimum routing cost for S . Then we have

$$R_T(S) \leq \frac{1}{k} \sum_{i=1}^k R_{T_i}(S) \quad (1)$$

Next we shall find an upper bound for $\sum_{i=1}^k R_{T_i}(S)$. First of all, we find an upper bound for $R_{T_i}(S)$. Let s and d be two vertices in V . Then we have the following inequality:

$$\begin{aligned}
R_{T_i}(S) &= \sum_{s \in S} \sum_{d \in V} D_{T_i}(s, d) \\
&\leq \sum_{s \in S} \sum_{d \in V} (D_{T_i}(s, s_i) + D_{T_i}(s_i, d)) \\
&\quad (\text{By triangle inequality}) \\
&\leq \sum_{s \in S} \sum_{d \in V} (\alpha D_G(s, s_i) + \alpha D_G(s_i, d)) \\
&\quad (\text{By Lemma 1}) \\
&= \alpha \sum_{s \in S} \sum_{d \in V-S} (D_G(s, s_i) + D_G(s_i, d)) + \\
&\quad \alpha \sum_{s \in S} \sum_{d \in S} (D_G(s, s_i) + D_G(s_i, d))
\end{aligned}$$

$$\begin{aligned}
&= (n-k)\alpha \sum_{d \in S} D_G(s_i, d) + k\alpha \sum_{d \in V-S} D_G(s_i, d) \\
&\quad + (k-1)\alpha \sum_{d \in S} D_G(s_i, d) \\
&\quad + (k-1)\alpha \sum_{d \in S} D_G(s_i, d) \\
&= k\alpha \sum_{d \in V-S} D_G(s_i, d) \\
&\quad + \alpha(n+k-2) \sum_{d \in S} D_G(s_i, d) \quad (2)
\end{aligned}$$

By the above inequality (2), we have

$$\begin{aligned}
&\sum_{i=1}^k R_{T_i}(S) \\
&\leq k\alpha \sum_{s \in S} \sum_{d \in V-S} D_G(s, d) + \\
&\quad \alpha(n+k-2) \sum_{s \in S} \sum_{d \in S} D_G(s, d) \\
&= k\alpha R_G(S) \\
&\quad + \alpha(n-2) \sum_{s \in S} \sum_{d \in S} D_G(s, d) \quad (3)
\end{aligned}$$

Let $d_1, d_2, \dots, d_{n-k}$ be the vertices in $V-S$, then we have the following two inequalities:

$$\begin{aligned}
&(n-k) \sum_{s \in S} \sum_{d \in S} D_G(s, d) \\
&= (n-k) \sum_{i=1}^k \sum_{j=1}^k D_G(s_i, s_j) \\
&= \sum_{i=1}^k \sum_{j=1}^{k-1} \sum_{m=1}^{n-k} D_G(s_i, s_{(i+j) \bmod k}) \\
&\leq \sum_{i=1}^k \sum_{j=1}^{k-1} \sum_{m=1}^{n-k} D_G(s_i, d_m) \\
&\quad + \sum_{i=1}^k \sum_{j=1}^{k-1} \sum_{m=1}^{n-k} D_G(s_{(i+j) \bmod k}, d_m) \\
&= \sum_{i=1}^k \sum_{j=1}^{k-1} \sum_{m=1}^{n-k} D_G(s_i, d_m)
\end{aligned}$$

Algorithm 1

Input: Metric graph $G = (V, E, w)$ with a set $S = \{s_1, s_2, \dots, s_k\} \subseteq V$ of source vertices, $\alpha > 1$, $\text{MST}(G)$, T_M , $\text{SPT}(G, s_i)$, $T_S^{s_i}$ for $1 \leq i \leq k$

Output: A spanning tree T of G

1. // use each s_k as source vertex respectively to run Find-Last
for $(i = 1 \text{ to } k)$
 $T_i = \text{Find-Last}(G, s_i, \alpha, T_M, T_S^{s_i});$
2. // output the spanning tree with smallest k -source routing cost
 $c = R_{T_1}(S)$, $j = 1$;
for $(i = 2 \text{ to } k)$
if $(c > R_{T_i}(S))$
 $c = R_{T_i}(S)$, $j = i$;
output T_j ;

Figure 1:

$$\begin{aligned}
& + \sum_{i=1}^k \sum_{j=1}^{k-1} \sum_{m=1}^{n-k} D_G(s_i, d_m) \\
& = 2(k-1) \sum_{i=1}^k \sum_{m=1}^{n-k} D_G(s_i, d_m) \\
& = 2(k-1) \sum_{s \in S} \sum_{d \in V-S} D_G(s, d) \quad (4)
\end{aligned}$$

$$\begin{aligned}
& (n+k-2) \sum_{s \in S} \sum_{d \in S} D_G(s, d) \\
& = 2(k-1) \sum_{s \in S} \sum_{d \in S} D_G(s, d) \\
& \quad + (n-k) \sum_{s \in S} \sum_{d \in S} D_G(s, d) \\
& \leq 2(k-1) \sum_{s \in S} \sum_{d \in S} D_G(s, d) \\
& \quad + 2(k-1) \sum_{s \in S} \sum_{d \in V-S} D_G(s, d) \\
& \quad \text{(By Inequality (4))} \\
& = 2(k-1) \sum_{s \in S} \sum_{d \in V} D_G(s, d) \\
& = 2(k-1) R_G(S) \quad (5)
\end{aligned}$$

Based upon Inequality (5), we can rewrite inequality (3) as follows:

$$\begin{aligned}
& \sum_{i=1}^k R_{T_i}(S) \\
& \leq k\alpha R_G(S) \\
& \quad + \alpha \left(\frac{2(k-1)(n-2)}{n+k-2} \right) R_G(S)
\end{aligned}$$

By the above inequality and $R_G(S) \leq R_{T_G}(S)$,

Inequality (1) can be rewritten as follows:

$$\begin{aligned}
R_T(S) & \leq \frac{1}{k} \sum_{i=1}^k R_{T_i}(S) \\
& \leq \alpha \left(1 + \frac{2(k-1)(n-2)}{k(n+k-2)} \right) R_{T_G}(S)
\end{aligned}$$

□

By Lemma 2 and Lemma 3, the main theorem of this section is as follows:

Theorem 1 Given a metric graph $G = (V, E, w)$ with a set of source vertices $S = \{s_1, s_2, \dots, s_k\} \in V$, an $\alpha > 1$ and a MST of G , $\text{SPT}(G, s_i)$ for $1 \leq i \leq k$, a spanning tree T of G can be found in $O(kn)$ time, where $n = |V|$ and $k = |S|$, such that

$$\begin{aligned}
(1) \quad C_T & \leq \left(1 + \frac{2}{\alpha-1} \right) C_{\text{MST}(G)} \\
(2) \quad R_T(S) & \leq \alpha \left(1 + \frac{2(k-1)(n-2)}{k(n+k-2)} \right) R_{T_G}(S)
\end{aligned}$$

4 Conclusion

We have propose an to construct a spanning tree T for a metric graph $G = (V, E, w)$ with and a source vertex set $S \subseteq V$, where V , E and w denote the vertex set, the edge set and the nonnegative weight function on edges respectively, such that the building cost of T is less than $1 + 2/(\alpha-1)$ times of that of a MST of G , and the k -source routing cost of T is less than $\alpha(1 + \frac{2(k-1)(n-2)}{k(n+k-2)})$ times of that of a k -MRCT of G with respect to S , where $k = |S|$, $n = |V|$ and $\alpha > 1$. The optimal algorithm for such problem on metric graphs is still open now.

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